

543 Readmissions Project Data Preparation
Report

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1 Overview

This report documents the steps taken following exploratory data analysis of `healthcare_readmissions_dataset_train.csv`. From EDA, the dataset had 7958 rows and 19 features (including the target).

2 Data Cleaning

2.1 Missing Data

Only features `number_of_prior_visits` and `medications_prescribed` contained missing values (314 (4%) and 657 (8%) respectively). Among all rows with missing values, only 21 lacked both features, so plainly removing all rows with missing values was ruled out. Imputing strategies were deferred until after feature engineering as transformations were anticipated.

2.2 Outlier Handling

During EDA, 80 rows were identified as outliers and removed for having impossible age values. BMI outliers were noted but not addressed.

Using a z -score threshold of $|z| > 3.5$, 7 rows were identified for having BMI values outside the plausible range. These rows were removed, reducing the dataset to 7,951 rows, from 8,038 original and 7,958 post-EDA.

2.3 Errors and Duplicates

No additional error correction or duplicate removal was necessary at this stage; both had been addressed during EDA.

3 Feature Engineering

3.1 Single Variable Transforms

3.1.1 `medications_prescribed` $\rightarrow$ `is_prescribed`

`medications_prescribed` was binarised into `is_prescribed` (1 if the original value was greater than zero, 0 otherwise). As there was a long tail of values and for simpler imputation. Missing values were treated as 0. Other encoding approaches might be looked at in the future (e.g. ordinal bins), but this was decided to be the baseline.

3.1.2 `length_of_stay` $\rightarrow$ `length_of_stay_score`

Continuous length of stay (days) was mapped to an ordinal risk score (1–7) following the LACE Index convention [1], a widely used readmission risk scoring system. The mapping is shown in Table 1.

Table 1: Length-of-Stay Score Encoding (LACE Index)

Length of Stay (days)	Score
≤ 1	1
2	2
3	3
4–6	4
7–14	5
> 14	7

The original `length_of_stay` column was dropped after encoding.

3.1.3 `age` $\rightarrow$ `age_group`

Raw age was binned into rough age groups to capture the non-linear relationship between age and readmission risk, as observed during EDA.

3.1.4 `number_of_prior_visits` (type conversion)

`number_of_prior_visits` was cast to `float` to allow downstream imputation and use in numeric models.

4 Feature Selection Decisions

The following variables were removed from the dataset:

- `weight_kg`: dropped because `adjusted_weight_kg` is highly correlated and is presumed to be the clinically adjusted version of the same measurement.
- `age`: replaced by `age_group` (see Section 3.1).
- `length_of_stay`: replaced by `length_of_stay_score` (see Section 3.1).
- `medications_prescribed`: replaced by `is_prescribed` (see Section 3.1).

Oversampling and standardization techniques were not applied at this stage, as more thorough experimentation seems needed to find the best approach for this dataset.

Imputing techniques were also kept as simple as possible for this baseline version. `number_of_prior_visits` was imputed to 0, but like medication imputing, will be explored further in future iterations.

5 Final Dataset Summary

`number_of_prior_visits` missing values were imputed to 0 as a conservative default.

Table 2: Dataset Info		
Column	Non-Null Count	Dtype
patient_id	7951	int64
gender	7951	str
ethnicity	7951	str
hospital_id	7951	str
height_m	7951	float64
smoker	7951	bool
bmi	7951	float64
adjusted_weight_kg	7951	float64
has_diabetes	7951	int64
has_hypertension	7951	int64
exercise_frequency	7951	str
diet_type	7951	str
number_of_prior_visits	7951	float64
type_of_treatment	7951	str
readmission_within_30_days	7951	int64
is_prescribed	7951	int64
length_of_stay_score	7951	int64
age_group	7951	category

The final preprocessed dataset contains **7,951 rows** and the columns listed in Table 2.

The file was saved to `data/interim/healthcare_readmissions_dataset_train_preprocessed.csv`.

References

- [1] van Walraven, C., Dhalla, I.A., Bell, C., et al. (2010). Derivation and validation of an index to predict early death or unplanned readmission after discharge from hospital to the community. *CMAJ*, 182(6), 551–557.